# Supplementary Material for "Real-Time Nodal Frequency-Secure AC Dispatch via Partially Input Convex Neural Networks"

Equation, table, and section numbers without the S prefix refer to the main paper; symbols not redefined here retain their main-text meaning.

## APPENDIX A
## SYSTEM CONSTANTS

Table SI lists the constants used in the dispatch-side constraints and the nodal dynamic model of Appendix B-C. The reported IEEE 118- and 300-bus studies use linear energy and reserve offers, whose coefficients are provided in the repository.

TABLE SI
SYSTEM AND NODAL-DYNAMIC-MODEL PARAMETERS.

| Symbol / item | Value | Meaning |
|---|---|---|
| $f_0$ | 50 Hz | Nominal frequency |
| $\overline{\Delta f}$ | 0.5 Hz | UFLS Stage-1 cap (49.5 Hz trigger), (5) |
| $T_d$ | 2.0 s | Slow SG/IBR reserve-delivery time |
| Governor | TGOV1 | SG governor and analogous GFM outer loop |
| $R_q^{\text{droop}}$ | 0.05 | SG/GFM droop coefficient (p.u. power on $P^B$ per p.u. speed) |
| $R_{\text{F}}^{\text{droop}}$ | 0.05 | GFL/BESS FFR droop coefficient (p.u. / p.u.) |
| $T_{1q}, T_{2q}, T_{3q}$ (SG) | 0.05/1.0/2.1 s | TGOV1 lead–lag constants |
| $T_{1q}, T_{2q}, T_{3q}$ (GFM) | 0.5/0/1.0 s | GFM outer-loop constants |
| $D_q$ | 2.5 | Effective damping coefficient applied to each swing node (p.u. power on $P^B$ per p.u. speed) |
| $H_{\text{SG}}^{\text{dev}}$ | 3.5 s | SG device-base inertia |
| $H_{\text{GFM}}^{\text{v}}$ | 2.0 s | GFM virtual inertia (conservative lower end) |
| $H_{\text{GFL}}^{\text{v}}$ | 0 | GFL carries *no* inertia credit |
| $\tau_{\text{IBR}}^{\text{F}}$ | 0.5 s | IBR fast-frequency-response outer-loop time constant |
| $\tau_{\text{BESS}}^{\text{F}}$ | 0.1 s | BESS FFR outer-loop time constant |
| $X'_{d,\text{SG}}, X'_{d,\text{GFM}}$ | 0.5/0.2 | Device-base transient coupling reactances (p.u.) |
| cond. retention | 0 / 0.3 | Condenser share: single-/multi-period study |

The damping sensitivity varies the common coefficient over $D_q \in [1, 5]$. The controller, inertia, damping, and reactance entries are representative device-class settings, with full per-unit parameters available in the repository.

## APPENDIX B
## LABEL GENERATION, SURROGATE TRAINING, AND THE NODAL FREQUENCY MODEL

### A. Plant-Cluster Label-Generation Protocol

*a) Contingency rule:* At each IBR penetration level, the model defends the loss of the fleet's *largest single-POI plant*: all IBR units behind one point of interconnection trip together with it as one event, so one plant is one $N-1$ element. Within each level, the *binding* plant is the one whose dispatched cluster output first exceeds its securable threshold, set by dispatch concentration rather than nameplate.

*b) Dispatch–engine consistency:* The SOCP and reference engine use the same cluster membership: (i) the SOCP contingency size is the sum of dispatched output over the co-located cluster, not a single unit; (ii) the post-contingency aggregates $H_c$ and $R_c$ exclude every co-located unit of the tripped plant, so no sibling unit is credited with inertia or reserve; (iii) the engine trips the full cluster member list, swing-participating and co-located grid-following units alike.

*c) Label grid:* Two label sets are generated with the nodal dynamic model of Appendix B-C under TGOV1 with $D$=2.5. The plant-cluster set carries 1,728 rows per penetration level (34/70/97%), spanning operating points around the dispatch distribution and a graded per-point de-load grid, and supervises the cluster-aware surrogate of Section V-B. The dispersed single-unit catalogue supplies the training and held-out points of the benchmark in Table I. The cluster-aware surrogate is fitted, conformally calibrated, and evaluated on mutually disjoint sets of operating points, with the closed-loop dispatches of the main text held out from both fitting and calibration.

### B. PICNN Training Protocol

The architecture of Section III-B [1] is trained with the Adam optimizer (learning rate $10^{-3}$, weight decay $10^{-4}$), a cosine-annealed schedule, batch size 256, a Huber loss with transition point 0.05 on the standardized target, and early stopping on validation. The deployed model is selected by validation RMSE.

### C. Nodal Post-Contingency Frequency Model

*a) Scope, sets, and dispatch interface:* The model specializes the nodal-frequency architecture of [2] to the SG, GFM/GFL IBR, and BESS models used here. It retains the electromechanical and outer-loop frequency states while neglecting faster converter inner-loop dynamics, following the usual time-scale separation in system-level frequency studies [3], [4]. The network is represented by the lossless, constant-voltage, small-angle specialization of the operating-point synchronization model in [2], [5].

For contingency $c$, let $\mathcal{D}_c^{\text{sw}}$ contain the surviving SGs, GFM IBRs, and retained synchronous condensers represented by swing states, and let $\mathcal{D}_c^{\text{F}}$ contain the surviving responsive GFL IBR and BESS FFR states. Indices $q$ and $j$ refer to these two sets, respectively. A zero award gives a zero response state.

With $N = |\mathcal{N}|$, $N_c^{\rm sw} = |\mathcal{D}_c^{\rm sw}|$, and $N_c^{\rm F} = |\mathcal{D}_c^{\rm F}|$, the dynamic state is

$$\boldsymbol{\xi}_c = [\boldsymbol{\delta}^\top, \boldsymbol{\omega}^\top, \boldsymbol{x}_1^\top, \boldsymbol{x}_2^\top, (\boldsymbol{p}^{\rm F})^\top]^\top. \tag{S1}$$

Here $\boldsymbol{\delta}, \boldsymbol{\omega}, \boldsymbol{x}_1, \boldsymbol{x}_2 \in \mathbb{R}^{N_c^{\rm sw}}$ are the internal angles, normalized speeds, and TGOV1 states, while $\boldsymbol{p}^{\rm F} \in \mathbb{R}^{N_c^{\rm F}}$ contains incremental nonswing responses. Dynamic powers are in p.u. on $P^B$, angles are in rad, and $f_0$ and $\omega_s = 2\pi f_0$ are the nominal frequency and angular speed. For a retained synchronous condenser, the two controller states are zero dummy states and remain zero.

Each operating point supplies the pre-trip active setpoints, the main-text SG/IBR awards $R_u$, the BESS awards $r_{b,t}^{\rm F}/P^B$, and the fixed BESS base injections $\bar{p}_{b,t}^{\rm ES}/P^B$. Let $\ell_{n,c}^0$ allocate the fixed non-dispatchable loss across buses, with $\sum_n \ell_{n,c}^0 = d_c^0$, and let $\mathcal{I}^{\rm GFL} \subset \mathcal{I}$ collect the grid-following IBRs. The post-trip physical-bus injection is

$$[\boldsymbol{p}_B^{(c)}]_n = \sum_{\substack{i \in \mathcal{I}^{\rm GFL} \setminus \mathcal{K}_c \\ n(i)=n}} P_i + \frac{1}{P^B} \sum_{\substack{b \in \mathcal{B} \\ n(b)=n}} \bar{p}_{b,t}^{\rm ES} - P_n^D - \ell_{n,c}^0. \tag{S2}$$

Thus positive entries denote net injections. Swing devices inject through their internal nodes in (S9) and therefore do not enter $\boldsymbol{p}_B^{(c)}$. The reported test systems use $d_c^0 = 0$. Each swing device $q$ has a corresponding main-text resource $u(q) \in \mathcal{U}$; a retained synchronous condenser enters $\mathcal{U}$ as a synchronous resource with $\underline{P}_u = \overline{P}_{u,t} = 0$. Then $p_q^{m,0} = P_{u(q)}$ and the pre-trip equilibrium satisfies $p_q^{e,0} = p_q^{m,0}$. Thus the same tripped set $\mathcal{K}_c$ determines the loss $d_c$, surviving inertia $H_c$, and surviving reserve $R_c$ in (1).

*b) Swing and response dynamics:* For a swing device $q \in \mathcal{D}_c^{\rm sw}$, let $H_q^{\rm dev}$, $S_q^{\rm base}$, and $D_q$ be its device-base inertia constant, MVA rating, and effective damping coefficient. Define

$$M_q = H_q^{\rm dev} S_q^{\rm base}/P^B, \qquad \Gamma_q = M_q/f_0. \tag{S3}$$

The coefficients $M_q$ and $\Gamma_q$ have units s and s/Hz, respectively, and the dispatch aggregate satisfies $H_c = \sum_{q \in \mathcal{D}_c^{\rm sw}} \Gamma_q$. A GFM unit uses its certified virtual-inertia setting for $H_q^{\rm dev}$; GFL IBRs and BESS units have no swing state and zero $\Gamma_u$. The swing equations are

$$\begin{aligned} \dot{\delta}_q &= \omega_s(\omega_q - 1), \\ 2M_q \dot{\omega}_q &= p_q^m - p_q^e - D_q(\omega_q - 1). \end{aligned} \tag{S4}$$

Here $p_q^m$ and $p_q^e$ are the mechanical or outer-loop command and electrical active power, respectively, in p.u. on $P^B$. The load-damping coefficient reported in the main text is represented by the common effective value $D_q = D$ at each swing node, following the effective-damping convention of low-order frequency-response models [3], [6].

Each responsive SG or GFM swing device uses the saturated TGOV1 lead–lag realization [7]

$$\begin{aligned} g_q &= {\rm sat}_{[\underline{p}_q^m, \overline{p}_q^m]}\left(p_q^{m,0} - \frac{\omega_q - 1}{R_q^{\rm droop}}\right), \\ T_{1q}\dot{x}_{1q} &= g_q - x_{1q}, \\ T_{3q}\dot{x}_{2q} &= (1 - a_q)x_{1q} - x_{2q}, \\ p_q^m &= a_q x_{1q} + x_{2q}. \end{aligned} \tag{S5}$$

The operator ${\rm sat}_{[a,b]}$ clips its argument to $[a, b]$, $R_q^{\rm droop}$ is the governor droop, $T_{1q}, T_{2q}, T_{3q}$ are the lead–lag time constants, and $a_q = T_{2q}/T_{3q}$. Both $R_q^{\rm droop}$ and $D_q$ are referred to the system base $P^B$, so a common setting gives every responsive swing device the same power–frequency gain. Here $p_q^{m,0}$ is the pre-contingency active-power command. For the underfrequency event, the award $R_q^{\rm aw}$ sets $\underline{p}_q^m = \max\{p_q^{m,0} - R_q^{\rm aw}, 0\}$ and $\overline{p}_q^m = p_q^{m,0} + R_q^{\rm aw}$; $R_q^{\rm aw}$ is the corresponding main-text reserve award $R_u$. Because $a_q \in [0, 1]$, the saturation confines $x_{1q}$ to $[\underline{p}_q^m, \overline{p}_q^m]$ and $x_{2q}$ to $(1 - a_q)[\underline{p}_q^m, \overline{p}_q^m]$ from their pre-trip values onward, so $p_q^m$ stays within the awarded band and the delivered response never exceeds $R_q^{\rm aw}$. A retained synchronous condenser has $p_q^{m,0} = R_q^{\rm aw} = 0$. The GFM realization uses the same equations with its virtual-inertia and outer-loop parameters, consistent with swing-based droop and virtual-inertia representations of grid-forming control [8]. The lower actuator limit is not credited as upward reserve in the adequacy constraint. For a nonresponsive swing device, $R_q^{\rm aw} = 0$ and $p_q^m$ remains at $p_q^{m,0}$.

The center-of-inertia (COI) frequency of the surviving swing set is

$$f_{\rm coi} = f_0 \left[1 + \frac{\sum_{q \in \mathcal{D}_c^{\rm sw}} M_q(\omega_q - 1)}{\sum_{q \in \mathcal{D}_c^{\rm sw}} M_q}\right]. \tag{S6}$$

The contingency catalogue ensures $\sum_{q \in \mathcal{D}_c^{\rm sw}} M_q > 0$. GFL and BESS responses are nonswing injections governed by

$$\tau_j^{\rm F} \dot{p}_j^{\rm F} = {\rm sat}_{[0, R_j^{\rm aw}]}\big(k_j[f_0 - f_{\rm coi}]\big) - p_j^{\rm F}, \qquad j \in \mathcal{D}_c^{\rm F}. \tag{S7}$$

Let $C_j^{\rm F}/P^B$ be the response-capacity base in p.u., obtained from the GFL converter rating or the BESS active-power rating. With the common FFR droop coefficient $R_{\rm F}^{\rm droop} = 0.05$, $k_j = C_j^{\rm F}/(P^B f_0 R_{\rm F}^{\rm droop})$ is the assigned power–frequency gain in p.u./Hz. The parameter $\tau_j^{\rm F}$ is the outer-loop time constant, and $R_j^{\rm aw}$ is in p.u. The award is $R_j^{\rm aw} = R_i$ for a GFL IBR $i$ and $R_j^{\rm aw} = r_{b,t}^{\rm F}/P^B$ for a BESS $b$. The analytical reference in Section III aggregates SG/IBR awards into $R_c$ with the common delivery time $T_d$, whereas the label engine uses (S5)–(S7); the resulting dynamic difference is included in the learned residual.

*c) Network linearization and reduction:* Let $\boldsymbol{B}^{\rm net} \in \mathbb{R}^{N \times N}$ be the positive physical-network susceptance Laplacian. An in-service branch $(n, m)$ with reactance $x_{nm}$ contributes $-1/x_{nm}$ to the two off-diagonal entries and $1/x_{nm}$ to the corresponding diagonal entries. These are the constant-voltage, small-angle limit of the operating-point synchronization coefficients of [5]. Let $\boldsymbol{E}_G \in \{0,1\}^{N \times N_c^{\rm sw}}$ and $\boldsymbol{E}_F \in \{0,1\}^{N \times N_c^{\rm F}}$ be the swing-device and nonswing-response incidence matrices. If $X_{d,q}^{\prime \rm dev}$ is the device-base transient reactance, its system-base value is $X'_{d,q} = X_{d,q}^{\prime \rm dev} P^B/S_q^{\rm base}$. Let $\boldsymbol{X}'_d = {\rm diag}(X'_{d,q})$. The physical-bus block and positive internal-node coupling are

$$\boldsymbol{B}_{BB} = \boldsymbol{B}^{\rm net} + \boldsymbol{E}_G \boldsymbol{X}_d'^{-1} \boldsymbol{E}_G^\top, \qquad \boldsymbol{B}_{BG}^+ = \boldsymbol{E}_G \boldsymbol{X}_d'^{-1}. \tag{S8}$$

The actual cross block of the augmented Laplacian is $\boldsymbol{L}_{BG} = -\boldsymbol{B}_{BG}^+$. This convention makes the post-trip algebraic and electrical-power equations

$$\boldsymbol{B}_{BB}\boldsymbol{\theta} = \boldsymbol{B}_{BG}^+\boldsymbol{\delta} + \boldsymbol{p}_B^{(c)} + \boldsymbol{E}_F \boldsymbol{p}^{\rm F},$$

$$p^e \quad = X_d'^{-1}(\delta - E_G^\top \theta), \tag{S9}$$

where $\theta \in \mathbb{R}^N$ contains physical-bus angles and $p^e \in \mathbb{R}^{N_c^{\text{sw}}}$ the swing-device electrical powers. The block $B_{BB}$ is the island susceptance Laplacian grounded through the transient reactances, and is nonsingular when each post-trip island contains a surviving internal-node coupling. Eliminating $\theta$ gives the Kron-reduced map [9]

$$p^e = K_\delta \delta + K_B p_B^{(c)} + K_F p^{\mathrm{F}}, \tag{S10}$$

with

$$\begin{aligned} K_\delta &= X_d'^{-1}[I - E_G^\top B_{BB}^{-1} B_{BG}^+], \\ K_B &= -X_d'^{-1} E_G^\top B_{BB}^{-1}, \qquad\qquad K_F = K_B E_F. \end{aligned} \tag{S11}$$

Equations (S4)–(S7) with (S10) form a closed state-space model for $\xi_c$. It is the saturated, resource-resolved counterpart of the linear nodal model obtained by network partition and Kron reduction in [2].

*d) All-bus frequency reconstruction:* Under the same constant-voltage and small-angle specialization, the frequency-formation map [10], [11] gives

$$\begin{aligned} f_{\text{bus}} &= f_0\big[\mathbf{1} + D_{\text{fdf}}(\omega - \mathbf{1})\big], \\ D_{\text{fdf}} &= B_{BB}^{-1} B_{BG}^+ = -B_{BB}^{-1} L_{BG}, \end{aligned} \tag{S12}$$

where $f_{\text{bus}} \in \mathbb{R}^N$ contains the frequencies of all physical buses and $\mathbf{1}$ is an all-ones vector of the required dimension. The two expressions for $D_{\text{fdf}}$ are identical under the positive-coupling convention above. Since $B^{\text{net}}\mathbf{1} = \mathbf{0}$ and each column of $E_G$ selects a single bus, $B_{BB}\mathbf{1} = E_G X_d'^{-1}\mathbf{1} = B_{BG}^+\mathbf{1}$ and hence $D_{\text{fdf}}\mathbf{1} = \mathbf{1}$, so a coherent swing-speed deviation is reproduced at every bus. Nonswing injections affect $f_{\text{bus}}$ through their network-coupled effect on $\omega$ in (S10).

*e) Contingency initialization and output metric:* Let $p_B^{(0)}$ denote (S2) on the intact network with $\mathcal{K}_c = \varnothing$ and $\ell_{n,c}^0 = 0$. The pre-trip bus and internal angles solve

$$\begin{bmatrix} B_{BB}^{(0)} & -B_{BG}^{+,(0)} \\ -(X_d'^{(0)})^{-1}(E_G^{(0)})^\top & (X_d'^{(0)})^{-1} \end{bmatrix} \begin{bmatrix} \theta^0 \\ \delta^0 \end{bmatrix} = \begin{bmatrix} p_B^{(0)} \\ p^{m,0} \end{bmatrix}, \tag{S13}$$

with one physical-bus angle fixed as the reference and $p^{m,0}$ collecting the commands of all pre-trip swing devices. At dynamic time $t' = 0$, every member of the plant cluster $\mathcal{K}_c$ is removed, including its base injection, inertia, and response award, and the matrices in (S8) are rebuilt. Surviving internal angles and controller states are continuous across the trip, with

$$\begin{aligned} &\delta(0^+) = \delta_{\text{surv}}^0, \quad \omega(0) = \mathbf{1}, \\ &x_1(0) = p_{\text{s}}^{m,0}, \quad x_2(0) = (\mathbf{1} - a) \odot p_{\text{s}}^{m,0}, \\ &p^{\mathrm{F}}(0) = \mathbf{0}. \end{aligned} \tag{S14}$$

Here $\delta_{\text{surv}}^0$ and $p_{\text{s}}^{m,0}$ are the surviving-device subvectors of $\delta^0$ and $p^{m,0}$, and $a = (a_q)_q$. At $\omega = \mathbf{1}$ these values give $g_q = p_q^{m,0}$, $\dot{x}_{1q} = \dot{x}_{2q} = 0$, and $p_q^m = p_q^{m,0} = p_q^{e,0}$, so the pre-trip state is an equilibrium of (S4)–(S5) and the response is driven solely by the removal of $\mathcal{K}_c$. The algebraic bus angles $\theta(0^+)$ are recomputed from (S9); they are not carried over from the pre-trip network.

Each response is integrated over $[0, 12]$ s with LSODA (rtol $= 10^{-7}$, atol $= 10^{-9}$, maximum step 0.02 s) and evaluated at 3000 uniformly spaced times. Internal machine nodes are excluded from $f_{\text{bus}}$. If $f_n^{(c)}(t')$ is its component for physical bus $n$, the metric used in the main text is

$$\begin{aligned} \Delta f_{n,c}^{\text{nad}} &= \max_{t' \in [0.15, 12]} \min_{s \in [t'-0.15, t']} [f_0 - f_n^{(c)}(s)], \\ \Delta f_c^{\text{wb}}(x) &= \max_{n \in \mathcal{N}} \Delta f_{n,c}^{\text{nad}}. \end{aligned} \tag{S15}$$

Thus the dispatch variables determine a unique plant-trip trajectory and worst-bus relay-window drop.

## D. Sensitivity

Halving the grid-forming virtual-inertia credit ($H_{\text{GFM}}^{\text{v}}$ : 2.0 $\to$ 1.0 s) leaves all three penetration levels secure: the worst-bus relay-window drop under the binding cluster $N-1$ moves from 0.481/0.432/0.385 Hz to 0.482/0.465/0.445 Hz at 34/70/97%, all below the 0.5 Hz cap with zero shed. Table SIII reports the companion sweep over the load-damping coefficient.

# Appendix C
# Fleet Summary, Sensitivity, and Computational Sweep

Table SII lists the IBR fleets at three penetration levels. Each plant aggregates co-located units at one point of interconnection (POI), and the binding plant is determined by dispatched cluster output rather than nameplate.

TABLE SII
IEEE 118-bus fleet summary across IBR penetration levels.

| IBR penetration | IBR cap. (MW) | IBR units | SGs online | Largest single-POI plants |
|---|---|---|---|---|
| 34% | 2196 | 4 | 54 | bus 69: 591 GFM + 591 GFL = 1182 MW |
| 70% | 4559 | 10 | 48 | + bus 66 (784), bus 100 (653) |
| 97% | 6336 | 18 | 40 | + bus 89 (637); remaining plants up to 509 |

Holding each dispatch fixed and varying only the common swing-node damping yields the sweep in Table SIII.

TABLE SIII
Damping sensitivity with fixed dispatches and TGOV1 retained.

| IBR penetration | Proposed-only secure for | COI-constrained shed |
|---|---|---|
| 34% | $D \in \{2.0, 2.5\}$ | 376.1 MW for $D \le 2.5$; zero for $D \ge 3.5$ |
| 70% | all $D \in [1, 5]$ | 345.9 $\to$ 13.3 MW as $D$ rises from 1 to 5 |
| 97% | $D \le 2.5$ | 196.8 $\to$ 0 MW as $D$ rises from 1 to 2.5 |

Table SIV reports median solution times over paired scenarios. The IEEE 118 mixed-integer linear program (MILP) encodes a capacity-matched deep neural network (DNN), whereas the IEEE 300 MILP is an exact big-M encoding of the PICNN used to check LP-block consistency.

TABLE SIV
MEDIAN SOLUTION TIMES FOR THE PROPOSED SOCP/LP EMBEDDING AND THE BIG-M MILP COMPARISONS.

| System | Events | SOCP/LP (s) | MILP (s) |
|---|---|---|---|
| IEEE 118 | 1 | 0.45 | 3.2 |
| | 2 | 0.28 | 4.4 |
| | 4 | 0.41 | 17.3 |
| | 8 | 0.70 | 191 |
| IEEE 300 | 5 | 0.04 | 0.44 |
| | 10 | 0.08 | 1.21 |
| | 25 | 0.40 | 3.27 |

The eight-contingency IEEE 118 MILP uses 1,536 binaries (192 per contingency). For IEEE 300, presolve removes stably inactive ReLU neurons, leaving 3,637 binaries with 25 contingencies; the SOCP/LP formulation is unchanged. A fixed model-assembly cost accounts for most of the subsecond IEEE 118 times, so they are not monotone in the event count.

## REFERENCES


[1] B. Amos, L. Xu, and J. Z. Kolter, "Input convex neural networks," in *Proc. 34th Int. Conf. Mach. Learn. (ICML)*, Sydney, Australia, Aug. 2017, pp. 146–155.

[2] Z. Pan, T. Wu, K. Xie, and T. Zhao, "Nodal frequency nadir chance-constrained economic dispatch considering contingency frequency service from hydrogen electrolysis," *IEEE Trans. Smart Grid*, 2026, early access, doi: 10.1109/TSG.2026.3684658.

[3] Q. Shi, F. Li *et al.*, "Analytical method to aggregate multi-machine SFR model with applications in power system dynamic studies," *IEEE Transactions on Power Systems*, vol. 33, no. 6, pp. 6355–6367, Nov. 2018.

[4] Z. Yang, M. Zhan, D. Liu *et al.*, "Small-signal synchronous stability of a new-generation power system with 100% renewable energy," *IEEE Transactions on Power Systems*, vol. 38, no. 5, pp. 4269–4280, Sep. 2023.

[5] R. Prasad and N. P. Padhy, "Spatiotemporal pattern analysis of nodal inertia using convex combination of walks of generator nodes in varying inertia system," *IEEE Transactions on Power Systems*, vol. 38, no. 5, pp. 4476–4488, Sep. 2023.

[6] P. M. Anderson and M. Mirheydar, "A low-order system frequency response model," *IEEE Trans. Power Syst.*, vol. 5, no. 3, pp. 720–729, Aug. 1990.

[7] IEEE PES Task Force on Turbine-Governor Modeling, "Dynamic Models for Turbine-Governors in Power System Studies," IEEE Power and Energy Society, Tech. Rep. PES-TR1, Jan. 2013.

[8] W. Du, F. K. Tuffner, K. P. Schneider, R. H. Lasseter, J. Xie, Z. Chen, and B. P. Bhattarai, "Modeling of grid-forming and grid-following inverters for dynamic simulation of large-scale distribution systems," *IEEE Trans. Power Del.*, vol. 36, no. 4, pp. 2035–2045, Aug. 2021.

[9] F. Dörfler and F. Bullo, "Kron reduction of graphs with applications to electrical networks," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 60, no. 1, pp. 150–163, Jan. 2013.

[10] F. Milano and Á. Ortega, "Frequency divider," *IEEE Transactions on Power Systems*, vol. 32, no. 2, pp. 1493–1501, 2017.

[11] J. Liu, C. Wang, J. Zhao, and T. Bi, "RoCoF constrained unit commitment considering spatial difference in frequency dynamics," *IEEE Trans. Power Syst.*, vol. 39, no. 1, pp. 1111–1125, Jan. 2024.